

UBoot64 v2



Figure 1: Logo

Boot menu for C64 Ultimate enabled devices — v2 (Oscar64 rebuild)

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Figure 2: Splash



Figure 3: Startup

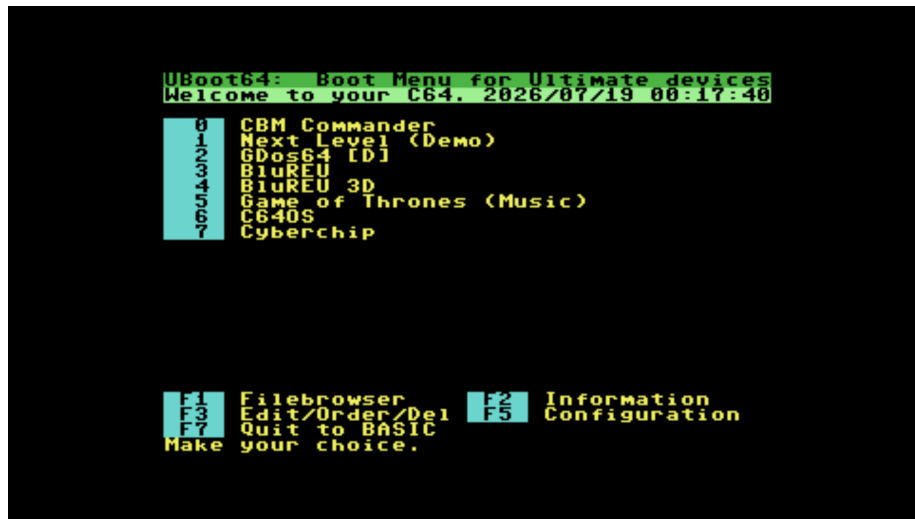


Figure 4: Main menu

Link to latest build:

Latest build

Version 2.2.0 - 20260816:

- USB port auto-reroute: if a slot's disk image or REU file was saved against one USB port (USB0/1/2) and the stick has since been moved to another, UBoot64 now detects this at boot time, automatically retries on the other USB ports, and reroutes to it for that boot if found — no need to re-edit the slot. Checks the actual file, not just the folder, so it correctly skips a USB stick that happens to share a folder name but not the file itself. See USB port auto-reroute.
- If the stick isn't found on any USB port, boot now stops with a clear "insert USB stick" message and lets you retry after reinserting it, instead of a generic mount error — press F7 to give up and return to BASIC instead.

Version 2.1.1 - 20260719:

- Configuration and slot files can now live on an SD card as well as USB storage. On startup, UBoot64 looks in this order: SD, USB0, USB1, USB2, and uses the first location where files already exist; a fresh install creates them on the first of those that's actually present, in the same order. Previously only a single USB drive was checked, and an SD-only setup would fail to start.
- The v1-to-v2 upgrade tool (`uboot_upd12.prg`) uses the same search order, so an upgrade always lands on the same device the cartridge will look for it on.

Version 2.1.0 - 20260719:

- Default boot slot with configurable auto-boot timeout. Set a slot as default in the Edit/Re-order/Delete menu (F6), set the timeout in the Configuration menu (F4, Off/1/3/5/10 sec). If configured, the boot menu shows a countdown screen before the normal menu; any keypress cancels it, letting the timeout expire boots the default slot automatically.
- Menu slots can now hold just a disk mount and/or a BASIC command, with no program to launch — useful for “auto-mount a disk on power-on” style slots. Create one via the Filebrowser’s **A/B** mount actions, or via **F2** (Edit command) on an empty slot in the edit menu.
- Adding a drive A image to a slot that already launches a program now warns first, since it replaces the disk the program expects to boot from.
- Fixed: auto-boot could hang in an endless loop when the default slot was at a letter position (A–H).
- Fixed: startup could fail to detect an installed REU (worked around a code generation issue in the current Oscar64 compiler).
- Fixed: the filebrowser’s side status panel could show garbled/misplaced text under certain UCI mode and mount-state combinations (same underlying compiler issue).
- Various other stability and bug fixes.

Version 2.0.1 - 202604230

- Fix for freeze at startup (error in cartridge init code)

Version 2.0.0 - 20260429:

- Complete rebuild using the Oscar64 cross-compiler (replaces cc65)
- Directory listings stored in REU memory, enabling:
 - much larger directory support
 - long filename support: maximum filename size now 50
 - long pathname support: maximum pathsize now 255. NB: The maximum is for the complete path, not for every directoryname in the path separately.
- Configurable colour scheme (F5 → F6 in configuration menu)
- Verbose or silent startup option (F5 → F2 in configuration menu)
- Splash screen in Information menu
- Configuration upgrade tool (`uboot_upd12.prg`) included for users upgrading from v1
- Various stability and bug fixes

Version 0.91 - 20230922:

- First public alpha (v1, cc65 build)

Building from source

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Build tooling

Tool	Purpose	Install
Oscar64	C cross-compiler targeting the 6502	Build from source; see the Oscar64 repo for instructions
zip	Bundles the release ZIP	<code>sudo apt install zip</code>
wput	FTP upload for <code>make deploy</code>	<code>sudo apt install wput</code>
curl	Reachability check before deploy	<code>sudo apt install curl</code>
pandoc	Regenerates <code>README.pdf</code> from <code>README.md</code> (optional — <code>make all</code> warns and skips if absent)	<code>sudo apt install pandoc texlive-xetex</code>

Deployment configuration

`make deploy` uploads the built cartridge and upgrade tool straight to your Ultimate device over FTP. The device IP is kept out of git in a local `.env` file:

```
# .env (gitignored, never committed)
ULTIP1 = 192.168.1.xx
```

Optionally override the USB slot (defaults to `usb0`):

```
ULTUSB = usb1
```

Make targets

Target	Effect
<code>make all</code>	Builds <code>uboot64.crt</code> , <code>uboot_upd12.prg</code> , regenerates <code>README.pdf</code> , and bundles the release ZIP — all into <code>build/</code>
<code>make clean</code>	Removes everything in <code>build/</code>
<code>make docs</code>	Regenerates <code>README.pdf</code> only
<code>make check-deploy</code>	Pings the configured Ultimate device without deploying
<code>make deploy</code>	Rebuilds if needed, then uploads <code>uboot64.crt</code> and <code>uboot_upd12.prg</code> via FTP (requires <code>.env</code>)

Instructions

Prerequisites

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- UltimateII+ (U2+) cartridge installed on a real C64, or an Ultimate 64 / Commodore 64 Ultimate
- For Gideon's Logic products: Firmware at version 3.4 or higher (to have access to the UCI DRVINFO command — firmware page)
- For Commodore Ultimate products: Supported starting from initial firmware
- RAM Expansion Unit (REU), minimum 128 KB, needs to be enabled
- Ultimate Command Interface needs to be enabled
- USB storage (up to 3 drives) or an SD card, for the configuration files

Installation

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- Download the ZIP file with the latest build
- In the ZIP file you will find these instructions, a `.crt` file (the cartridge image) and a `uboot_upd12.prg` file (configuration upgrade tool for v1 users). Unzip contents to a temporary location.
- Transfer the `.crt` file to the `/Flash/Carts` directory on your Ultimate device. Either transfer via FTP, or place it on the USB storage, browse to it in the UI filebrowser, press **C= + C** to copy, navigate to `/Flash/Carts` and paste with **C= + V**.
- Select the `.crt` file as the cartridge to start automatically: in the Ultimate UI press **F2**, navigate to **Cartridge and ROM Settings**, press **RETURN** twice to edit the **Cartridge** option, and select `uboot64.crt` from the drop-down.
- In the same **Cartridge and ROM Settings** menu, ensure that RAM Expansion Memory (REU) is enabled. Size does not matter as even the smallest possible size of 128KB will be sufficient.
- In the same **Cartridge and ROM Settings** menu, ensure that the Ultimate Command Interface is enabled.
- Press **STOP** to leave the configuration menu and **RETURN** to confirm saving to flash.
- Power cycle your device. UBoot64 should now start automatically.
- Stop autostarting by deselecting the cartridge via the same procedure and choosing **None** or another image.



Figure 5: Ultimate UI

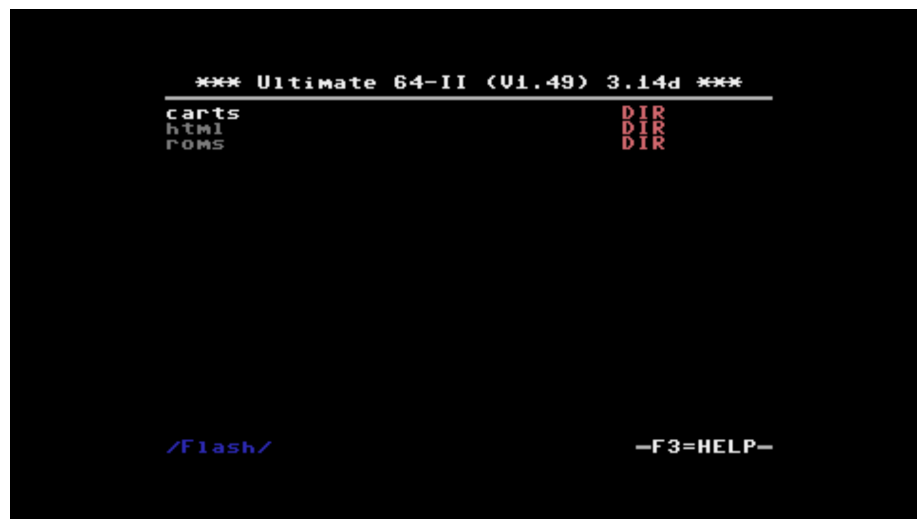


Figure 6: Ultimate Flash dir

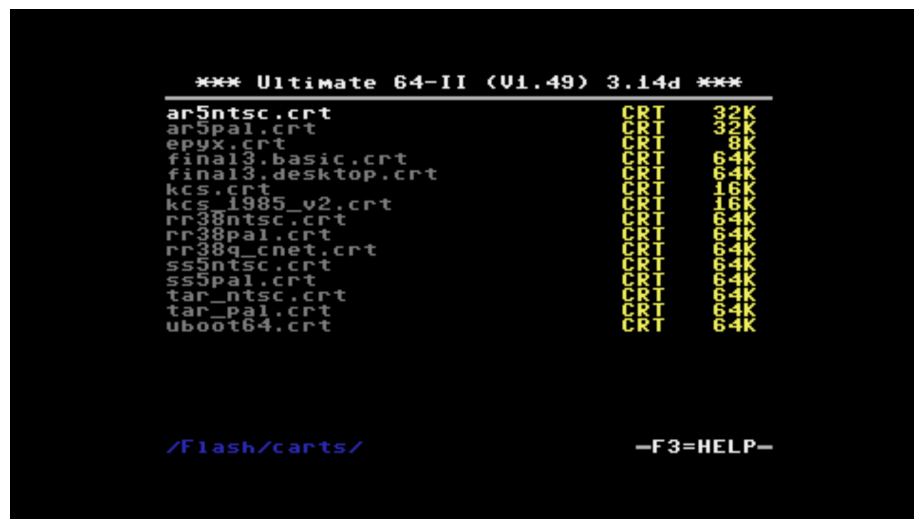


Figure 7: Ultimate Carts dir

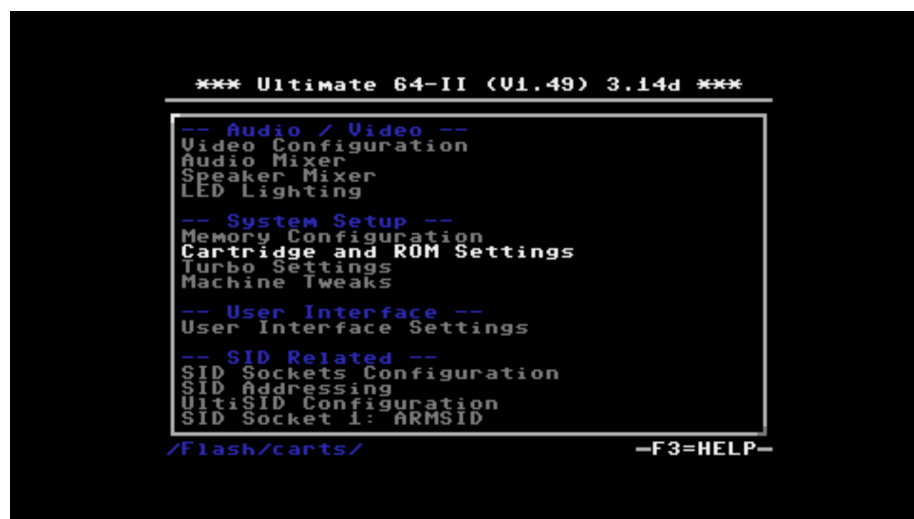


Figure 8: Ultimate UI cart settings

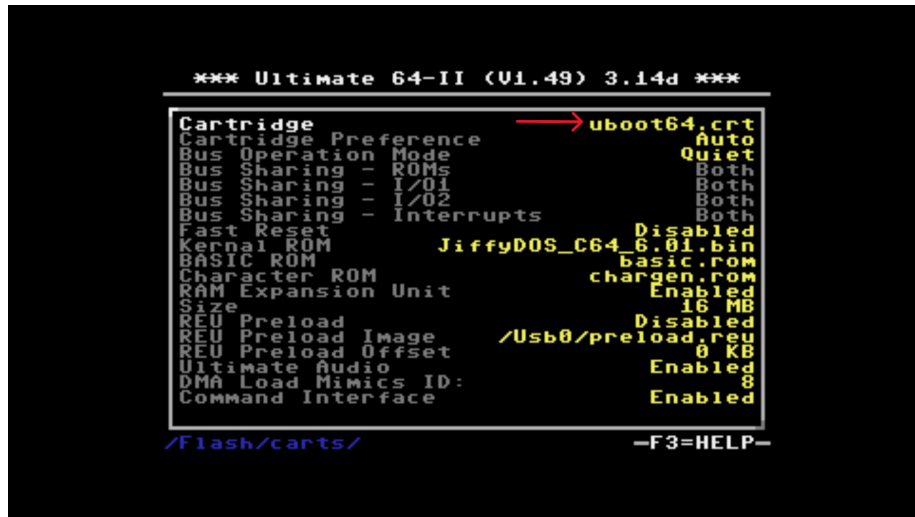


Figure 9: Ultimate UI select cart

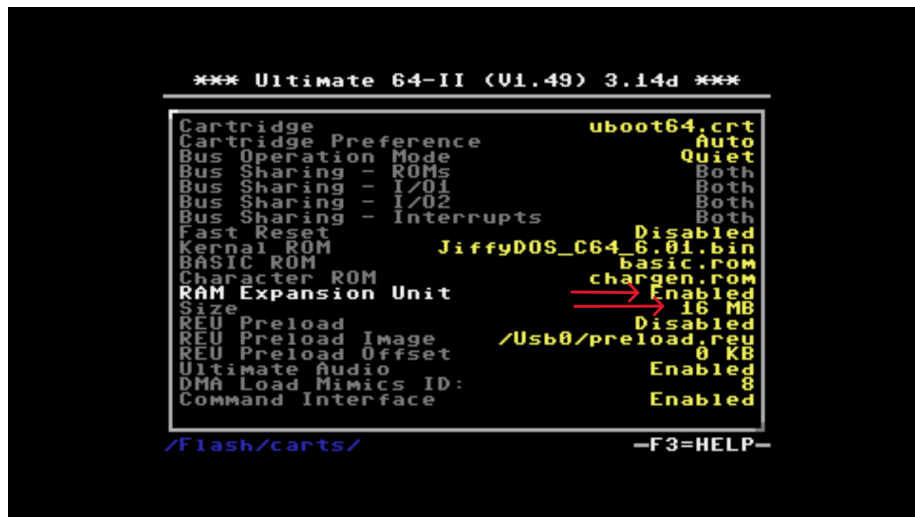


Figure 10: Ultimate UI enable REU

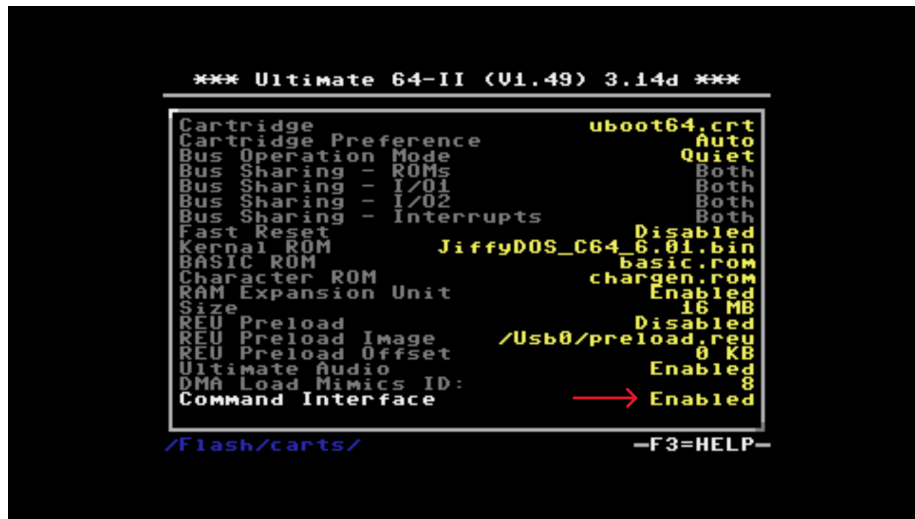


Figure 11: Ultimate UI enable UCI

Upgrading from v1

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If you have an existing v1 configuration (slot and config files), you must run the upgrade tool before using v2 for the first time. The slot file format changed between v1 and v2.

- Transfer uboot_upd12.prg to your USB storage.
- Run uboot_upd12.prg from the Ultimate UI or a BASIC prompt before starting UBoot64 v2.
- The tool reads your existing v1 configuration files and rewrites them in the v2 format.
- After the upgrade tool completes successfully, start UBoot64 v2 normally.

If you do not run the upgrade tool, UBoot64 v2 will detect the old format and exit with an error asking you to run the upgrade tool first.

First run

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- At first run, no configuration file is present yet, so UBoot64 creates configuration files with an empty start menu and default settings.
- UBoot64 looks for existing configuration files in this order: SD card, then USB0, USB1, USB2. The first one where they're found is used; if none have them, they're created on the first of those that's actually present (same order). The files are DMBCFG.CFG (configuration data) and DMBSLT.CFG

- (menu slot data), always at the root of that device.
- On startup a splash screen is shown briefly. Press any key to proceed to the main menu.
- The empty start menu looks like this:

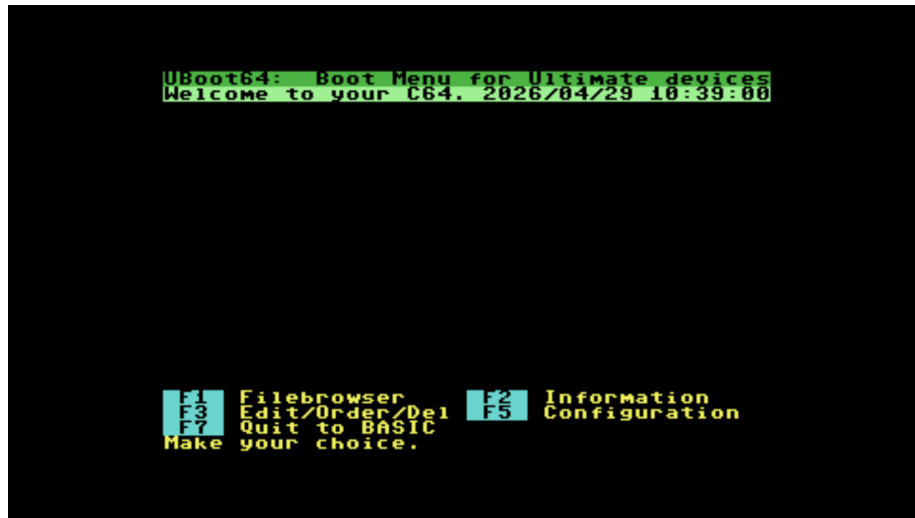


Figure 12: Empty boot menu on first run

Add start options via the Filebrowser

- Start options can be added to menu slots **0–9** and **A–Z** (18 slots total) via the Filebrowser.
- Press **F1** to open the filebrowser:
- The browser starts in **UCI mode** — browsing via the Ultimate Command Interface on the native filesystem of your Ultimate device.
- Press **F3** to switch to **IEC mode** for drives connected to the IEC bus. Use **+** / **-** to select the device number.
- Press **F1** to refresh the directory if needed.
- In IEC mode, press **D** to start a **directory trace** from the root, needed to record the full path for a menu slot. The TRACE toggle in the lower right switches to ON.
- **,1 Load** (toggle with **1**): loads with `LOAD"NAME",8` (off) or `LOAD"NAME",8,1` (on).
- **Demo mode** (toggle with **O**): powers down all drives except ID 8 on start.
- Navigate with **cursor keys** (UP/DOWN in dir, ENTER/RIGHT to enter dir or image, DEL/LEFT to go up).



Figure 13: File browser in UCI mode



Figure 14: File browser in IEC mode



Figure 15: File browser status toggles panel in UCI mode



Figure 16: Status toggles panel with directory trace enabled

- From UCI mode, entering a disk image (.Dxx extension) mounts it on drive A and switches to IEC mode — “Inside mount” appears in the menu.
- Press **ENTER** on a program file to select it for a menu slot. In UCI mode the associated drive A disk image is automatically included.
- Press **A** or **B** on a disk image (.Dxx) to select it as the additional image to mount on drive A resp. B when starting from that slot.
- Press **M** on a program file to select it to be run from the disk image already mounted on drive A.
- To add a REU file: navigate to it and press **ENTER**, then select the target slot, choose REU size with + / -, and confirm with **ENTER**.

Note: The REU file must be in the same filepath as the drive A disk image if one is present.

- After selecting what to place in a slot, you are shown this screen to pick the slot position:
- If the slot is already filled, confirmation is requested. Choose **Y** to proceed or **N** to cancel.
- Press **0-9** or **A-Z** to choose the slot, enter a name and press **ENTER**.
- Repeat until all desired slots are filled.
- Important notes:



Figure 17: File browser showing inside a mounted disk image



Figure 18: Inside mount indicator in the file browser sidebar

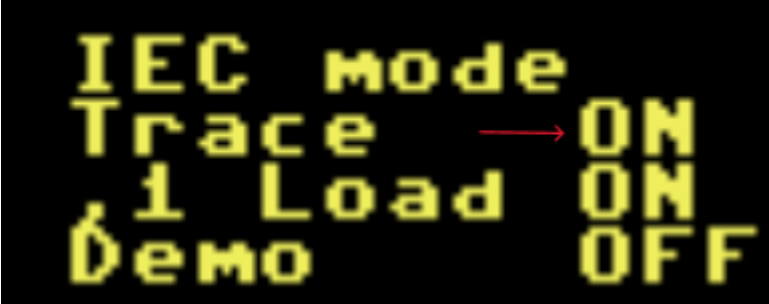


Figure 19: Status toggles panel with directory trace enabled



Figure 20: Pick slot screen to assign the selected item to a menu slot

- For technical reasons, loose .PRG files in UCI mode can not be added to the start menu. Either add them via the SoftIEC in IEC mode, or use the Ultimate native UI for launching these applications. Reason is that launching a PRG from an UCI path is not supported via the Ultimate Command Interface (yet)
- If you want to add a program, disk images and a REU image to the same slot, this needs to be done in separate steps for each file or image to add. Just choose the same slot again each time.

Mount/command-only slots A menu slot does not need a program to launch. Pressing **A** or **B** on a disk image without ever selecting a program to run creates a slot that only mounts that disk and then lands at the BASIC READY. prompt on boot — the classic “auto-mount a disk on power-on” setup, especially combined with a default boot slot and timeout.

You can also create a slot that runs only a BASIC command, with no disk mount and no program: in the edit menu (**F3**) press **F2** (Edit command) and pick an **empty** slot — you’ll be prompted for a name, then the command.

These combine freely: adding a drive B image or a REU preload to a slot that already launches a program keeps the program intact (useful for programs that expect a second disk or preloaded data). Adding a drive A image to a slot that already launches a program **replaces** the disk the program expects to boot from, so you’ll be asked to confirm first — if you just want an extra mount alongside an existing program, add it as drive B instead, or use a fresh slot for a drive-A-only mount.



Figure 21: Drive A mount conflict warning

F1: Filebrowse menu

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The filebrowser is based on and inspired by the DraBrowse program from <https://github.com/doj/dracopy>

Key reference

Key	Function
F1	Read / refresh directory
F3	Toggle between UCI and IEC modes
+	Increase device number (IEC mode only)
-	Decrease device number (IEC mode only)
RETURN	Enter directory / run selected program (IEC mode) / select for slot
CURSOR RIGHT	Enter directory
DEL	Go to parent directory
CURSOR LEFT	Go to parent directory
↑	Go to root directory
T	Go to the first item in the directory
E	Go to the last item in the directory
P	Page down in the directory
U	Page up in the directory
Cursor keys	Navigate in the directory

Key	Function
D	Toggle Dirtrace (IEC mode only): traces directory movements from root
A	Select highlighted image (.Dxx) to be mounted on drive A
B	Select highlighted image (.Dxx) to be mounted on drive B
M	Select highlighted program to run from the disk image on drive A
1	Toggle ,1 load
O	Toggle demo mode
Q or F7	Quit to main menu

UCI mode and IEC mode The file browser opens in **UCI mode** by default. Press **F3** to toggle between modes. The current mode is shown in the lower right of the sidebar.

UCI mode browses the native filesystem of the Ultimate device (USB storage) directly via the Ultimate Command Interface. It supports long filenames, full directory paths, and can enter disk images (.D64, .G64, .D71, .G71, .DNP) as if they were directories.

IEC mode reads directories from a physical or emulated IEC bus device (disk drives on the serial bus). Press **+** or **-** to cycle through available active device numbers. In IEC mode, filenames follow CBM conventions and directory listings are read via the KERNAL. File type labels (PRG, SEQ, DEL, REL, CBM, DIR) are shown next to each entry.

Directory navigation Use **CURSOR UP** / **CURSOR DOWN** to move the highlight one entry at a time. Use **P** (page down) and **U** (page up) to jump a full screen at a time. **T** jumps to the first entry in the directory and **E** jumps to the last.

To enter a subdirectory or disk image, press **RETURN** or **CURSOR RIGHT** on it. To go back up to the parent directory, press **DEL** or **CURSOR LEFT**. Press **↑** (the up-arrow key, not cursor up) to jump directly to the root of the current file system.

Press **F1** at any time to re-read and refresh the current directory listing.

Directory trace (IEC mode) Because IEC drives do not report their full path, pressing **D** activates **directory trace** mode: UBoot64 records each directory you navigate into, building the full path from root. The TRACE indicator in the sidebar switches to **ON**. Navigate from the root directory with trace active before selecting a program, so the complete path can be stored in the menu slot.



Figure 22: File browser showing directory listing

Without trace active in IEC mode, only the filename is stored; the path will be empty and the program can only be found if it is in the root directory of the drive.



Figure 23: Status toggles panel with directory trace enabled

Press **D** again to turn trace off and reset the recorded path depth.

Toggles: ,1 Load and Demo mode The sidebar shows two additional toggles that affect how a program is launched when selected for a slot.

,1 Load (toggle with **1**): controls the BASIC LOAD command used at boot time. - **OFF**: LOAD"FILENAME",8 — loads the program to the address in the file header (normal for most games and programs). - **ON**: LOAD"FILENAME",8,1 — loads to the absolute address stored in the file. Needed for some utilities and programs that require loading at a fixed memory address.

Demo mode (toggle with **O**): when ON, all Ultimate emulated drives except the one on IEC ID 8 are powered off before the program starts. Many demos and some games require only one drive to be active. Enable this toggle when adding demos to ensure reliable playback on systems with multiple emulated drives.



Figure 24: File browser status toggles panel in UCI mode

Both toggle states are stored in the menu slot when you confirm the selection.

Selecting items for menu slots After navigating to the desired file or image, use the appropriate key for the type of item:

Selecting a program to run (RETURN or M): - In IEC mode, pressing **RETURN** on a .PRG file without trace active runs it immediately (direct launch, no slot storage). - In IEC mode with trace active, pressing **RETURN** on a .PRG stores it in a slot — the recorded path and filename are saved. - In UCI mode, pressing **RETURN** on a .PRG file does nothing unless first a disk image was entered. If a disk image was entered before (since mount), the program stores it in a selected slot and also automatically stores that disk image as the drive A image. - Pressing **M** instead of **RETURN** stores the program to run from the disk image already mounted on drive A (IEC inside-mount mode). Use this when the program must be started from within the mounted disk.

Entering and using a disk image (RETURN on .Dxx file in UCI mode): Pressing **RETURN** on a disk image file in UCI mode mounts it on drive A and switches to IEC mode, showing “Inside mount” in the sidebar. You can then browse the disk contents and press **RETURN** on a program to store both the program and the disk image in a slot.

Adding an additional drive B image (B): In UCI mode, pressing **B** on a disk image (.Dxx) stores it as the drive B image for the next slot you choose. This is useful when a program needs two drives simultaneously.

Adding a drive A image without entering it (A): Pressing **A** on a disk image in UCI mode stores it as the drive A image for the next slot, without mounting or entering the image. Use this when you want to add an image to a slot separately from the program selection step.

Selecting a REU image (RETURN on .REU file): Navigate to a .REU file and press **RETURN**. You are then prompted to choose the REU size with + / - before confirming. The REU image is preloaded into REU memory before the program in the slot starts.

After any selection, you are taken to the slot picker screen to choose which of the 18 menu slots (0–9, A–H) the item should be stored in.

Limitations in UCI mode Loose .PRG files on the native UCI file system cannot be added to a menu slot for direct launch. The Ultimate Command Interface does not (yet) support launching a program directly from a USB path. Options: - Use IEC mode via SoftIEC to access the file through a CBM device number. - Enter a disk image in UCI mode first, then select the program from within the mounted image. - Use the Ultimate's own UI for running programs directly from USB.

F2: Information

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Shows first the splash screen, and after pressing any key, the information and credits screen. Press any key to return to the main menu.



Figure 25: Splash

F3: Edit / re-order / delete

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Rename, re-order, edit commands for, or delete menu slots.

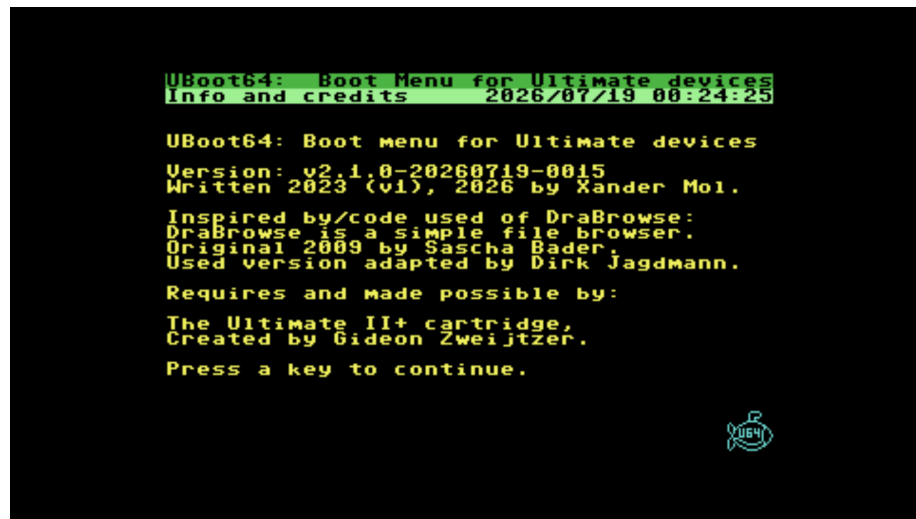


Figure 26: Information and credits screen



Figure 27: Edit, re-order and delete slots menu

- **F1** — Rename a slot. Choose slot (**0–9 / A–Z**), enter new name, press **ENTER**.



Figure 28: Rename slot screen

- **F2** — Add or edit a user-defined BASIC command to execute before the program starts.
- **F3** — Re-order slots. Choose slot to move (**0–9 / A–Z**), highlighted in white. Press **UP / DOWN** to move. Confirm with **ENTER**, cancel with **F7**.
- **F5** — Delete a slot. Choose slot (**0–9 / A–Z**), confirm with **Y** or cancel with **N**.
- **F6** — Set or clear the default boot slot. Choose a slot; slots currently marked default are shown with a **[D]** tag in the list. Choosing the slot that already has **[D]** clears it; choosing another slot moves the default there (only one slot can be default at a time). See Default boot slot and auto-boot timeout.
- **F7** — Return to main menu. Changes are saved at this point.

Default boot slot and auto-boot timeout

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You can mark one menu slot as the default boot target and have it launch automatically after a countdown, without needing to press a key — useful for auto-mounting a disk (or launching a program) on power-on.

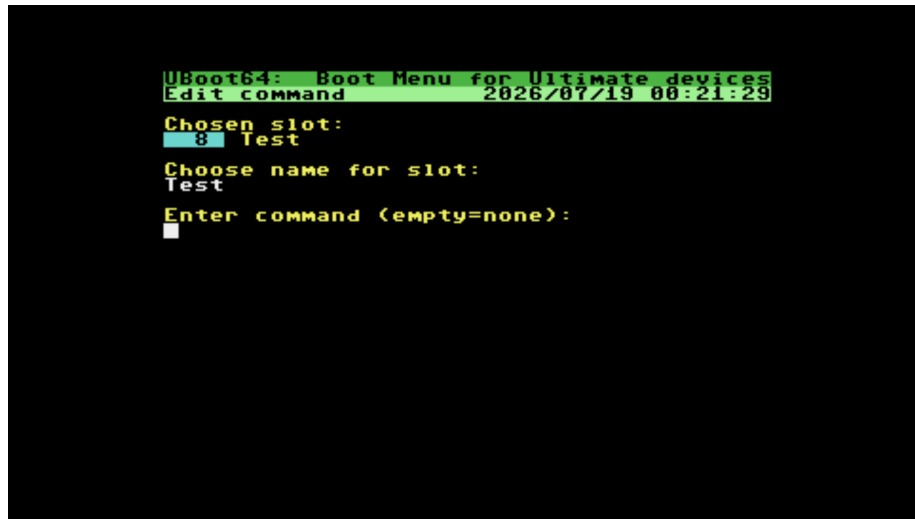


Figure 29: Edit user-defined BASIC command screen

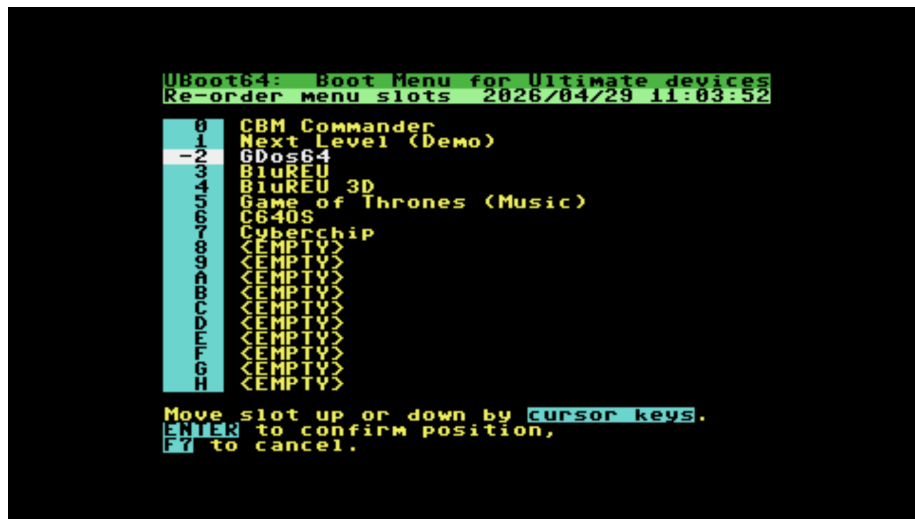


Figure 30: Re-order slots screen



Figure 31: Delete slot confirmation screen

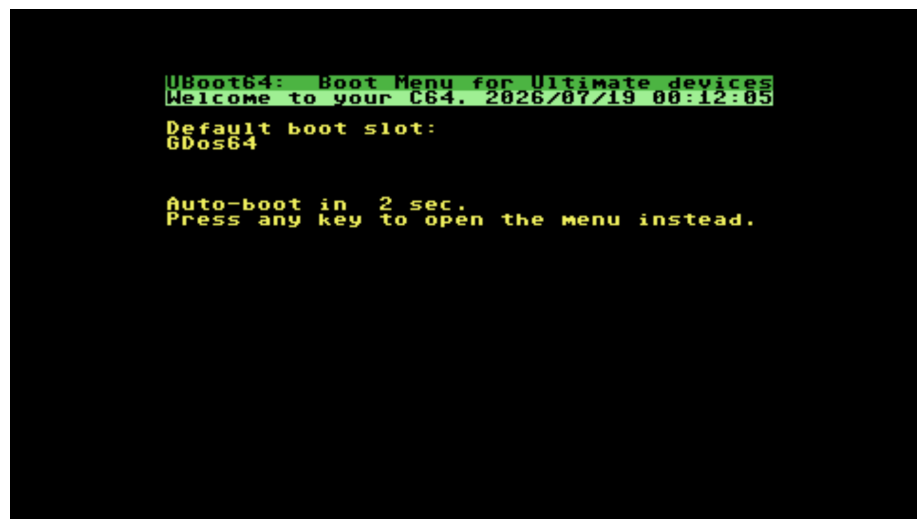


Figure 32: Auto-boot countdown screen

- Set the default slot: main menu → **F3** → **F6**, then pick a slot. Slots currently marked default show a [D] tag wherever slots are listed.

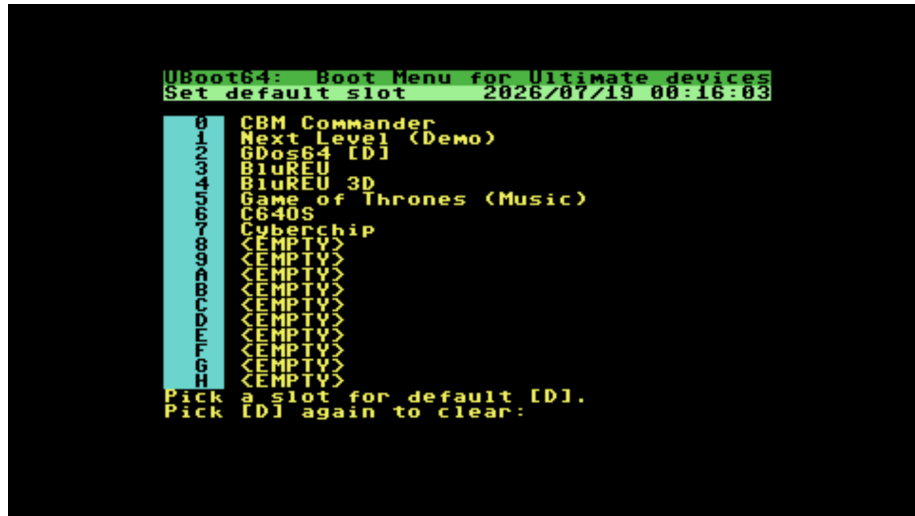


Figure 33: Set default boot slot screen

- Set the timeout: main menu → **F5** → **F4** to cycle Off / 1 / 3 / 5 / 10 seconds. Default is **Off**, so this has no effect until both a default slot and a timeout are configured.
- When both are set, the boot menu shows a dedicated countdown screen (rather than the full menu) with the default slot’s name and remaining seconds. Pressing **any key** cancels the countdown and opens the normal menu — the keypress itself is not treated as a selection. If the countdown reaches zero, the default slot boots automatically, exactly as if you had pressed its key.
- The default slot follows the slot’s content through renames and re-ordering. Deleting the slot (or picking it again via **F6**) clears the default marker.
- A default slot does not need to launch a program — see mount/command-only slots below for the classic “auto-mount a D64 on power-on” use case.

USB port auto-reroute

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A menu slot’s disk image (drive A / B) and REU file are stored with the USB port they were on at the time (e.g. /usb0/...). If you later plug that stick into a different USB port, the slot would previously fail to boot.

UBoot64 now detects this automatically at boot time: if the stored path and file aren’t found there, it retries on the other USB ports (USB0/1/2). If found,

it shows a brief “Rerouted to ...” message and continues booting from there — this only affects the current boot and is not saved back to the slot, so the slot still uses its originally configured port next time (letting you use multiple sticks across different slots as before). This checks the actual file, not just the folder — if two sticks happen to share a folder name (e.g. both have a **games** folder) but only one has the specific file the slot needs, UBoot64 correctly finds and boots from the right one instead of stopping on the wrong stick.

If the file isn’t found on any USB port, boot stops with a message asking you to insert the stick — press any key to retry once it’s plugged back in, or **F7** to give up and return to BASIC (useful if you meant to boot something else entirely).

This only applies to disk image and REU paths (which use USB port addressing via the Ultimate Command Interface). It does not apply to programs launched directly from an IEC device without a disk mount, or to the SD card slot, since neither uses a swappable USB port.

F5: Configuration

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Configure NTP time synchronisation, startup verbosity, and the UI colour scheme.

After pressing **F5**:



Figure 34: Configuration menu showing NTP time sync and other settings

The screen shows current settings and allows editing:

- **F1** — Toggle NTP time sync on or off. Default: enabled.

- **F2** — Toggle verbose or silent startup. Verbose shows detailed feedback during startup; silent shows only a progress indicator. Default: verbose.
- **F3** — Edit the time offset to UTC in seconds. Automated daylight saving adjustment is not provided; adjust manually when needed. Examples: CET = 3600, CEST = 7200. See <https://www.timeanddate.com/time/zones/> for all offsets (multiply hours by 3600). Default: 0 (UTC).
- **F4** — Cycle the auto-boot timeout: Off, 1, 3, 5, or 10 seconds. Default: Off. See Default boot slot and auto-boot timeout.
- **F5** — Edit the NTP server hostname. Default: pool.ntp.org.
- **F6** — Edit the UI colour scheme. Use **CURSOR UP** / **DOWN** to select a colour element, **CURSOR LEFT** / **RIGHT** to change its colour value (0–15). **DEL** reverts to the saved colours. **F7** returns to the configuration menu. Changes are saved when you exit the configuration menu.

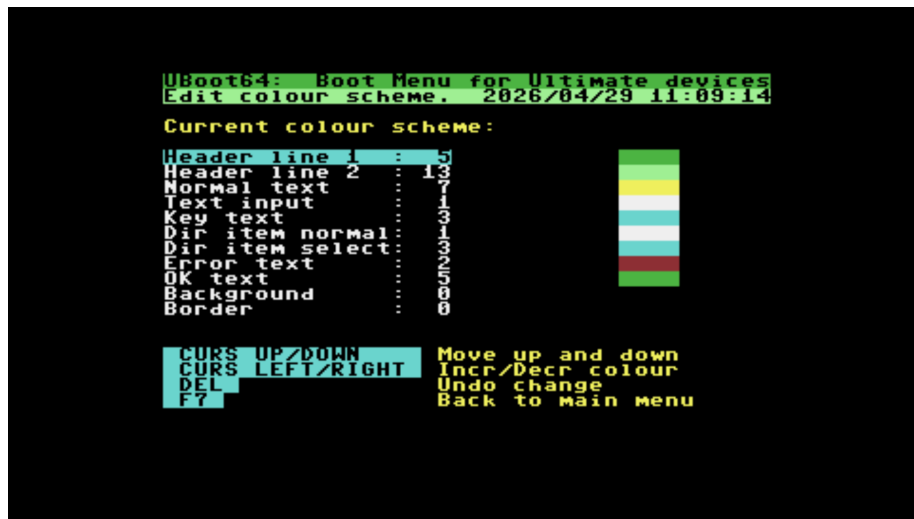


Figure 35: UI colour scheme editor

- **F7** — Return to main menu. Changes are saved.

F7: Quit to BASIC

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Exit the boot menu to the BASIC Ready prompt. Memory is erased on exit.

Credits

UBoot64 v2

Boot menu for C64 Ultimate enabled devices

Written in 2023 (v1), 2026 (v2) by Xander Mol <https://github.com/xahmol/UBoot64-v2> <https://www.idreamtin8bits.com/>

Inspired by and code used from DraBrowse: DraBrowse (db*) is a simple file browser. Originally created 2009 by Sascha Bader. Used version adapted by Dirk Jagdmann (doj) <https://github.com/doj/dracopy>

Most of code and functionality ported from: DMBoot 128: Device Manager Boot Menu for the Commodore 128 Written in 2020-2023 by Xander Mol <https://github.com/xahmol/DMBoot>

Built using: - Oscar64 cross-compiler by drmortalwombat <https://github.com/drmortalwombat/oscar64>

Additionally uses code from: - Ultimate 64/II+ Command Library Scott Hutter, Francesco Sblendorio <https://github.com/xlar54/ultimateii-dos-lib>
- ntp2ultimate by MaxPlap <https://github.com/MaxPlap/ntp2ultimate>
Time via NTP code - EPOCH-to-time-date-converter by sidsingh78 https://github.com/sidsingh78/EPOCH-to-time-date-converter/blob/master/epoch_conv.c
- petcom version 1.00 by Craig Bruce, 18-May-1995 Convert from PETSCII to ASCII, or vice-versa. https://codebase64.org/doku.php?id=source_conversion

Requires and made possible by the Ultimate II+ cartridge, Created by Gideon Zweijtzer <https://ultimate64.com/>

Bart van Leeuwen: For suggesting the default boot slot with configurable auto-boot timeout feature.

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